FOR SERVICE PERSONNEL ONLY

HITACHI

HITACHI SPLIT UNIT AIR CONDITIONER INSTALLATION MANUAL

INDOOR UNIT / OUTDOOR UNIT
RAS-D10EX / RAC-D10EX
RAS-D14EX / RAC-D14EX

HFC R410A

- Carefully read through the procedures of proper installation before starting installation work.
- The sales agent should inform customers regarding the correct operation of installation.

Tools Needed For Installation Work

(Mark ◉ is exclusive use tool for R410A) • ⊕⊖ Screwdriver • Measuring Tape • Knife • Saw • ø65mm Power Drill • Hexagonal Wrench Key (4mm) • Wrench (14,17,19, 22mm) ◉ Gas leakage Detector • Pipe Cutter • Putty • Vinyl Tape • Pliers • Flare Tool ◉ Vacuum Pump Adapter ◉ Manifold Valve ◉ Charge Hose ◉ Vacuum Pump

SAFETY PRECAUTION

- Read the safety precautions carefully before operating the unit.
- The contents of this section are vital to ensure safety. Please pay special attention to the following sign.

⚠ WARNING .... **Incorrect methods of installation may cause death or serious injury.**
⚠ CAUTION .... **Improper installation may result in serious consequence.**
Make sure to connect earth line.
This sign in the figures indicates prohibition.

Be sure that the unit operates in proper condition after installation. Explain to customer the proper operation and maintenance of the unit as described in the user's guide. Ask customers to keep this installation manual together with the instruction manual.

⚠ WARNING

- Please request your sales agent or qualified technician to install your unit. Water leakage, short circuit or fire may occur if you do the installation work yourself.
- Please observe the installation stated in the installation manual during the process of installation. Improper installation may cause water leakage, electric shock and fire.
- Make sure that the units are mounted at locations which are able to provide full support to the weight of the units. If not, the units may collapse and impose danger.
- Observe the rules and regulations of the electrical installation and the methods described in the installation manual when dealing with the electrical work. Use cables which are approved official in your country. Be sure to use the specified circuit. A short circuit and fire may occur due to the use of low quality wire or improper work.
- Be sure to use the specified cables for connecting the indoor and outdoor units. Please ensure that the connections are tight after the conductors of the wire are inserted into the terminals to prevent the external force is being applied to the connection section of the terminal base. Improper insertion and loose contact may cause over-heating and fire.
- Please use the specified components for installation work. Otherwise, the unit may collapse or water leakage, electric shock, fire or stronger vibration may occur.
- Be sure to use the specified piping set for R410A. Otherwise, this may result in broken copper pipes or faults.
- When installing or transferring an air conditioner to another location, make sure that air other than the specified refrigerant (R410A) does not enter the refrigeration cycle. If other air should enter, the pressure level of the refrigeration cycle may increase abnormally which could result in a rupture and injury.
- Be sure to ventilate fully if a refrigerant gas leak while at work. If the refrigerant gas comes into contact with fire, a poisonous gas may occur.
- After completion of installation work, check to make sure that there is no refrigeration gas leakage. If the refrigerant gas leaks into the room, coming into contact with fire in the fan-driven heater, space heater, etc., a poisonous gas may occur.
- Unauthorized modifications to the air conditioner may be dangerous. If a breakdown occurs please call a qualified air conditioner technician or electrician. Improper repairs may result in water leakage, electric shock and fire, etc.
- Be sure to connect the earth line from the power supply wire to the outdoor unit and between the outdoor and indoor unit. Do not connect the earth line to the gas tube, water pipe, lighting rod or the earth line of the telephone unit. Improper earthing may cause electric shocks.
- When finishing the refrigerant collection (pumping down), stop the compressor and then remove the coolant pipe. If you remove the refrigerant pipe while the compressor is operating and the service valve is released, air is sucked and a pressure in the freezing cycle system will build up steeply, causing an explosion or injury.
- When installing the unit, be sure to install the refrigerant pipe before starting the compressor. If the refrigerant pipe is not installed and the compressor is operated with the service valve released, air is sucked and the pressure level of the refrigeration cycle may increase abnormally which could result in a rupture and injury.

⚠ CAUTION

- A circuit breaker must be installed in the house distribution box for the direct connected power supply cables to the outdoor unit. In case of other installations a main switch with a contact gap or more than 3mm has to be installed. Without a circuit breaker, the danger of electric shock exists.
- Do not install the unit near a location where there is flammable gas. The outdoor unit may catch fire if flammable gas leaks around it.
- Please ensure smooth flow of water when installing the drain hose. Improper installing may wet your furniture.
- An IEC approved power cord should be used. Power cord type: NYM.

THE CHOICE OF MOUNTING SITE (Please note the following matters and obtain permission from customer before installation.)

INDOOR UNIT

⚠ WARNING
- The unit should be mounted at stable, non-vibratory location which can provide full support to the unit.

⚠ CAUTION
- No nearby heat source and no obstruction near the air outlet is allowed.
- The clearance distances from top, right and left are specified in figure below.
- The location must be convenient for water drainage and pipe connection with the Outdoor unit.
- To avoid interference from noise, please place the unit and its remote controller at least 1m from the radio and television.
- To avoid any error in signal transmission from the remote controller, please put the controller far away from high-frequency machines and high-power wireless systems.

OUTDOOR UNIT

⚠ WARNING
- The Outdoor unit must be mounted at a location which can support heavy weight. Otherwise, noise and vibration will increase.

⚠ CAUTION
- Do not expose the unit under direct sunshine or rain. Besides, ventilation must be good and clear of obstruction.
- The air blown out of the unit should not point directly to animals or plants.
- The clearances of the unit from top, left, right and front are specified in figure below. At least two of the above sides must be open air.
- The installation height should be at least 2,300 mm or more from the floor.
- Be sure that the hot air blown out of the unit and noise do not disturb the neighbourhood.
- Do not install at a location where there is flammable gas, steam, oil and smoke.
- The location must be convenient for water drainage.
- Place the Outdoor unit and its connecting cord at least 1 m away from the antenna or signal line of television, radio or telephone. This is to avoid noise interference.

No.	Item	Qty
①	Hanger	1
②	Screw for Hanger (4.1 × 32)	5
③	AAA size Battery	2
④	Remote Controller Mounting Screw (3.5 × 15)	1
⑤	Holder	1
⑥	Remote Controller	1
⑦	Insulation Sheet	3
⑧	Bush	1
⑨	Drain Pipe	1
⑩	Switch Cover	1
⑪	Purifying Filter	1
⑫	Net	1
⑬	Hood	1
⑭	Nano-Titanium Deodorizing/Dust Collecting Fresh Air In Filter	1
⑮	Air Duct for Piping Connection from left	1
⑯	Joint	1
⑰	Elbow	1

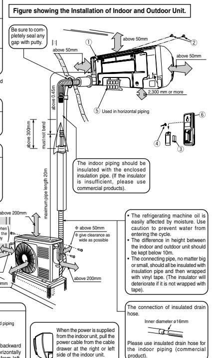
No.	Item	Qty
⑱	Two Way Air Exchange Hose (A) (ø25 × 2000)	1
⑲	Two Way Air Exchange Hose (B) (ø20 × 700)	1
⑳	Cushion	1

The components of ⑧ & ⑨ are included in the package of the outdoor unit.

The Length of Indoor Unit Connecting Cord: above 0.9m, above 1.6m

Dimension of Mounting Stand of the Outdoor Unit (unit : mm)

Figure showing the Installation of Indoor and Outdoor Unit.



- The refrigerating machine oil is easily affected by moisture. Use caution to prevent water from entering the cycle.
- The difference in height between the indoor and outdoor unit should be kept below 10m.
- The connecting pipe, no matter big or small, should all be insulated with insulation pipe and then wrapped with vinyl tape. (The insulator will deteriorate if it is not wrapped with tape).

Direction of Piping: Backward piping from left; Connection. There are 6 directions allowed, namely, backward piping, backward piping from left, horizontally piping from right, horizontally piping from left, vertically down from right, vertically down from left.

When the power is supplied from the indoor unit, pull the power cable from the cable drawer at the right or left side of the indoor unit. Cable drawer (right side)

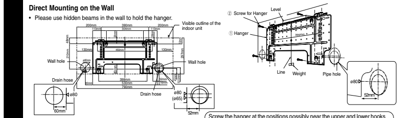
The connection of insulated drain hose. Inner diameter ø16mm. Please use insulated drain hose for the indoor piping (commercial product).

INDOOR UNIT

1 Installation of Wall Penetration and Installation of Protection Pipe

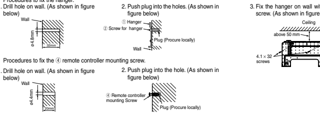
⚠ CAUTION
- The draining of the water container inside the Indoor unit can be done from the left. Therefore the hanger must be fixed horizontally or slightly tilted towards the side of drain hose. Otherwise, condensed water may overflow the water container.

Direct Mounting on the Wall
- Please use hidden beams in the wall to hold the hanger.

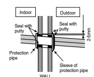


Screw the hanger at the positions possibly near the upper and lower hooks where the indoor unit is hung. Use 4 or more screws to fix the hanger.

Procedures of Installation and Precautions
- Procedures to fix the hanger.
  1. Drill hole on wall. (As shown in figure below)
  2. Push plug into the holes. (As shown in figure below)
  3. Fix the hanger on wall with 4.1 × 32 screw. (As shown in figure below)
- Procedures to fix the ④ remote controller mounting screw.
  1. Drill hole on wall. (As shown in figure below)
  2. Push plug into the hole. (As shown in figure below)



Wall Penetration and Installation of Protection Pipe
- Drill a ø65~80mm hole on wall which is slightly tilted towards the outdoor side. Drill the wall at a small angle. (ø70mm or more is recommended)
- Cut the protection pipe according to the wall thickness.
- Empty gap in the sleeve of protection pipe should be completely sealed with putty to avoid dripping of rain water into the room.
- Please ensure smooth flow of water when installing the drain hose. Improper installing may wet your furniture.
- An IEC approved power cord should be used. Power cord type: NYM.



⚠ WARNING
Be sure that the wire is not in contact with any metal in the wall. Please use the protection pipe as wire passing through the hollow part of the wall so as to prevent the possibility of damaged by mouse.
Unless it seals completely any air with high humidity flows from outdoor and any dew may drop.

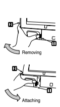
2 Installation of the Indoor Unit

Preparation of Installation

Remove the Front Panel
- To remove front panel, please refer to page 13 "Removing and Attaching the Front Panel".
- Make sure to use both hands to attach and remove the front panel.

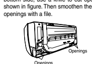
Remove the Top Grill
- To remove top grill, please refer to page 14 "Attaching the Top Grill".

Remove the Low Cover
- Pull at the 1 and 2 in the directions as shown by arrows to remove the low cover.
- To attach the lower cover, attach 2 first and then attach 1 by rotating it around the 2 as the fulcrum.



Wiring Connection
- Refer to page 10.

Cutting Low Cover Bush
- While installing the pipe from the right, left or bottom side, use a knife to cut openings as shown in figure. Then smoothen the edges of openings with a file.



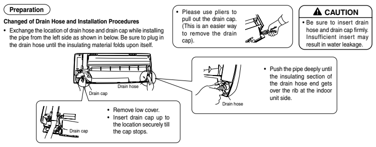
1 **PIPING FROM THE LEFT SIDE (BACKWARD, DOWNWARD, HORIZONTAL)**

Preparation

Changed of Drain Hose and Installation Procedures
- Exchange the location of drain hose and drain cap while installing the pipe from the left side as shown in below. Be sure to plug in the drain hose until the insulating material folds upon itself.

- Please use pliers to pull out the drain cap. (This is an easier way to remove the drain cap).

⚠ CAUTION
- Be sure to insert drain hose and drain cap firmly. Insufficient insert may result in water leakage.



- Remove low cover.
- Insert drain cap up to the location securely till the cap stops.
- Push the pipe deeply until the insulating section of the drain hose end gets over the rib at the indoor unit side.

2 **PIPING FROM THE RIGHT SIDE (BACKWARD, DOWNWARD, HORIZONTAL)**

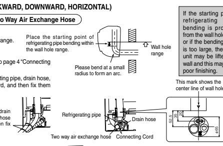
Preparation of Refrigerating Pipe and Two Way Air Exchange Hose

In case of backward piping

1 Form the refrigerating pipe within the wall hole range.

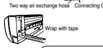
2 Connect the two way air exchange hose. (Refer to page 4 "Connecting and Fixing the Two Way Air Exchange Hose.")

3 Referring to the drawing below, join the refrigerating pipe, drain hose, two way air exchange hose and connecting cord, and then fix them temporarily with tape.



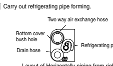
If the starting point of refrigerating pipe bending is projected from the wall hole range or if the bending radius is too large, the indoor unit may be lifted from wall and this may cause poor finishing.

4 Wrap the through-hole portion with tape.
- If the existing wall hole is deviated from the pipe position, cover the two way air exchange hose located at the back of the indoor unit with heat insulation sheet, with the joint of insulation sheet facing up.



In this case of horizontally piping and downward piping from right

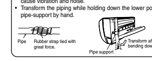
1 Carry out refrigerating pipe forming.



Layout of Horizontally piping from right.

2 Temporary join the refrigerating pipe, drain hose, two way air exchange hose and connecting cord with tape.

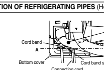
⚠ CAUTION
- The rubber strap used for fixing the insulator should not be tied with great force. Otherwise, this will damage heat insulation and causes water condensation.
- Please pull the lower part of the Indoor unit outwards to check if the unit is hooked onto the hanger. Improper installation may cause vibration and noise.
- Transform the piping while holding down the lower portion of pipe-support by hand.



INSTALLATION AFTER CONNECTION OF REFRIGERATING PIPES (Horizontally backward piping)

1 Connect the two way air exchange hose. (Refer to page 4 "Connecting and Fixing the Two Way Air Exchange Hose".)

2 Form the two way air exchange hose and connecting cord as shown in the drawing.



⚠ WARNING
- **Two way air exchange hose should be formed at higher level than Part A to avoid trapping, etc.**
  If two way air exchange hose is formed at lower level than Part A and trapping is caused, dew condensation around the heat insulation sheet may drip onto electrical parts and it may cause electrical shock or fire.

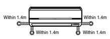
1 Two Way Air Exchange Hose Installation

The two way air exchange function is included in this model. Please follow the "two way air exchange hose" installation instructions when mounting.

Conditions when the two way air exchange should not be operated **(Please install only with the customer's consent)**.
- The two way air exchange should not be operated under the following conditions:
  (A) When the two way air exchange hose cannot be vented directly outside the room (mounting work specified for the embedded ducting).
  (B) When there is a source of a questionable odor, smoke, or oily smoke in the vicinity of the opening of the two way air exchange hose, even if the mounting work directly vents the two way air exchange hose outside the room.
- The fresh air in should not be operated (only exhaust can be operated) under the following conditions:
  (C) When there is space to vent the two way air exchange hose between the internal and external walls, which leads outside the room, even if the mounting work cannot directly vent the Two way air exchange hose outside the room (conventional construction methods).
- Please set the Two Way Air Exchange Selection Switch according to the mounting configuration (refer to "Setting the Two Way Air Exchange Selection Switch on page 12).

※ In the case of (A) or (C) in the above, two way air exchange can be operated if the two way air exchange hose is vented outside the room through a hole in the wall (refer to "End Treatment of Two Way Air Exchange Hose" on page 15).

[Dimensions when Installing included Fittings]



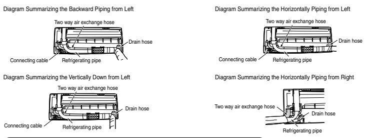
Dimensions when installing the fitting for ø25 two way air exchange hose (When piping connection is left, ø20 hose shall be attached first. The details shall conform to the table.)

UNUSED ACCESSORY

Unused accessories may become necessary in case of transfer of air conditioner. Request your customer to keep all the unused accessories together with this installation manual and instruction manual.

Use of Accessories Select (○) depending on the installation condition.

Part Name	⑱ ø25 Two way air exchange Hose (A)	⑲ ø20 Two way air exchange Hose (B)	⑮ Air duct for piping connection from left	⑫ Net	⑬ Hood	⑯ Joint	⑰ Elbow
Backward piping	○	–	–	○	○	–	–
Vertically down from right Horizontally piping from right	○	–	–	○	○	–	–
Backward piping from left Vertically down from left	First attach ⑲ ø20 hose add ⑯ joint or ⑰ elbow, then use this hose.	○	○	○	○	○	(○)
Horizontally piping from left	First attach ⑲ ø20 hose add ⑯ joint, then use this hose.	○	○	○	○	○	–
Remark					When Air Exchanger cover (HC-DS5) is not using.		



⚠ CAUTION
- **Ensure to incline the two way air exchange hose and install from the top towards downward.** Dew may form on the two way air exchange hose and there may be a possibility of water droplets forming.

< S655 : Ⓑ >

2 Length Adjustment of the Two Way Air Exchange Hose

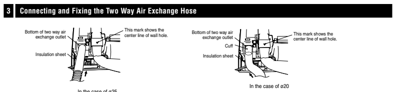
This model is supplied with two hoses. Details shall conform to the table in page 3.
Please ensure that the entire length of the two way air exchange hose is within 3m.

Please adjust the length of the two way air exchange hose by joining or cutting hoses, as deemed necessary.

Target hose length = Distance between the centre of the hole in the wall from the opening of the two way air exchange unit + Thickness of the wall + Over 230mm This will change according to the installation outside the room

Please shorten the length of the hose as much as possible to prevent the degradation of the two way air exchange operating performance.

3 Connecting and Fixing the Two Way Air Exchange Hose



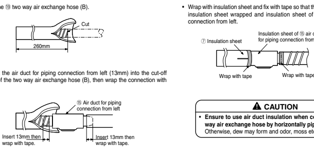
In the case of ø25 — In the case of ø20

- Please insert two way air exchange hose until the end of the outlet.

4 Installation of Air Duct for Piping Connection from Left

Only horizontally piping, except for the case where refrigerating pipe is connected after the indoor unit is hung on the wall.

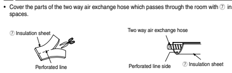
- Cut the ⑲ two way air exchange hose (B).
- Insert the air duct for piping connection from left (13mm) into the cut-off part of the two way air exchange hose (B), then wrap the connection with tape.
- Wrap with insulation sheet and fix with tape so that there is no gap between insulation sheet wrapped and insulation sheet of ⑮ air duct for piping connection from left.



⚠ CAUTION
- **Ensure to use air duct insulation when connecting two way air exchange hose by horizontally piping from left.** Otherwise, dew may form and odor, moss etc may form.

5 Heat Insulation Tape of the Two Way Air Exchange Hose

- Cover the parts of the two way air exchange hose which passes through the room with ⑦ insulation sheet for hoses and wrap with tape to ensure there are no spaces.



⚠ CAUTION
- **Ensure that the two way air exchange hose which passes through the room is covered with insulation sheet for hoses.** Dew may form on the two way air exchange hose and there may be a possibility of water droplets forming.

3 Cut the heat insulation sheet refrigerating pipe aligning to the ⑦ insulation sheet of the pipe and fix them temporarily with tape.

4 Wrap the refrigerating pipe and two way air exchange hose with ⑦ insulation sheet and fix it with tape. Tape must not be overtightened. (Refer to "Heat insulation and finish of the piping" on Page 11.)

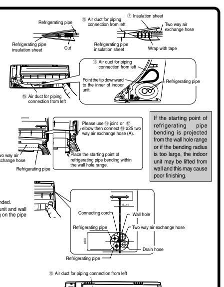
- Place the tip of air duct for horizontally backward piping in the direction shown in the drawing on the right.

5 Wrap two ways air exchange hose with ⑦ insulation sheet until wall hole.

6 Form the refrigerating pipe according to the wall hole position. Especially in case of horizontally backward piping, follow the instruction below to carry out accurate forming.

<Forming of refrigerating pipe for horizontally backward piping>
(1) Referring to the drawing on the right, start bending the refrigerating piping within wall hole range.
(2) When forming, bend the refrigerating pipe with the smallest radius as shown the drawing on the right. (Finishing will be better if the center of the forming is moved a little to the right.)

※ Wall hole of ø80 is recommended.
※ The gap between the indoor unit and wall surface may be big depending on the pipe forming.



If the starting point of refrigerating pipe bending is projected from the wall hole range or if the bending radius is too large, the indoor unit may be lifted from wall and this may cause poor finishing.

7 Form the connecting cord and refrigerating pipe and put them in the space on the lower side of the rear surface of the indoor unit. Fix them with the holder.

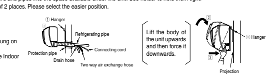
8 Join refrigerating pipe, two way air exchange hose, connecting cord and drain hose temporarily with tape and wrap the through-hole portion with tape.

9 Attach the ⑳ cushion as shown in the drawing.

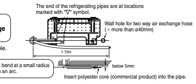
THE CONNECTION OF REFRIGERATING PIPE DURING THE INSTALLATION OF INDOOR UNIT
- The refrigerating pipe should be adjusted to fit into the hole on the wall and then ready for further connection.
- The terminals of 2 connected pipes must be covered with insulator used for terminal connection. Then the pipes are wrapped with insulation pipe.
- Connect the connecting cord after removing low cover. (Refer to "CONNECTION OF POWER CORD" on page 10.)
- After adjustment, fit the connecting cord and pipes into the space available under the unit. Use holder to hold them tight.
- Holder can be attached at the either of 2 places. Please select the easier position.

Installation

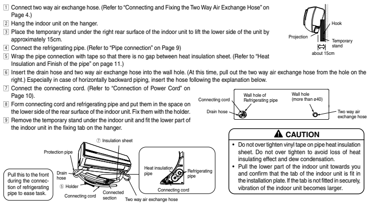
1 Insert the pipe through the wall hole.
2 The upper part of the Indoor unit is hung on the hanger.
3 The projection at the lower part of the Indoor unit is hooked onto the hanger.



If refrigerating pipe is connected at the back of the indoor unit, make the wall hole for two way air exchange hose on the right
- Insert refrigerating pipes and connecting cord through the wall hole.



1 Connect two way air exchange hose. (Refer to "Connecting and Fixing the Two Way Air Exchange Hose" on Page 4.)
2 Hang the indoor unit on the hanger.
3 Place the temporary stand under the right rear surface of the indoor unit to lift the lower side of the unit by approximately 15cm.
4 Connect the refrigerating pipe. (Refer to "Pipe connection" on Page 9)
5 Wrap the pipe connection with tape so that there is no gap between heat insulation sheet. (Refer to "Heat Insulation and Finish of the pipe" on page 11.)
6 Insert the drain hose and two way air exchange hose into the wall hole. (At this time, pull out the two way air exchange hose from the hole on the right.) Especially in case of horizontally backward piping, insert the hose following the explanation below.
7 Connect the connecting cord. (Refer to "Connection of Power Cord" on Page 10).
8 Form connecting cord and refrigerating pipe and put them in the space on the lower side of the rear surface of the indoor unit. Fix them with the holder.
9 Remove the temporary stand under the indoor unit and fit the lower part of the indoor unit in the fixing tab on the hanger.

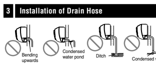


⚠ CAUTION
- Do not over tighten vinyl tape on pipe heat insulation sheet. Do not over tighten to avoid loss of heat insulating effect and dew condensation.
- Pull the lower part of the indoor unit towards you and confirm that the tab of the indoor unit is fit in the installation plate. If the tab is not fitted in securely, vibration of the indoor unit becomes larger.

How to Remove Indoor Unit
- Pull up the [PULL] sections at the bottom of the indoor unit and pull the bottom plate towards you. Then the claws are released from the mounting plate. (The [PULL] sections are indicated by 2 arrows in the right figure.)

[PULL] mark positions

3 Installation of Drain Hose

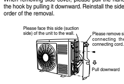


⚠ CAUTION
Be sure that the drain hose is not loosely connected bend or proper condition like left figure.

⚠ CAUTION
You are free to choose the side (left or right) for the installation of drain hose. Please ensure the smooth flow of condensed water of the Indoor unit during installation. (Careless may result in water leakage.)

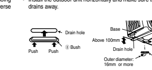
OUTDOOR UNIT

- Please mount the Outdoor unit on stable ground to prevent vibration and increase of noise level.
- Decide the location for piping after sorting out the different types of pipe available.
- When removing side cover, please pull the handle after undoing the hook by pulling it downward. Reinstall the side cover in reverse order of the removal.



Condensed Water Disposal of Outdoor Unit
- There is holes on the base of Outdoor unit for condensed water to exhaust.
- In order to flow condensed water to the drain, the unit is installed on a stand or a block so that the unit is 100mm above the ground as shown figure. Join the drain pipe to one hole.

- Cover the drain hole with a bush. To install the bush, put it on the drain hole as shown in the figure and press the both sides of the bush to fit into the hole. After installation, check whether the drain pipe and bush cling to the base firmly.
- Install the outdoor unit horizontally and make sure that condensate drains away.



In case of using in chilly area

Especially, in case that there are many snows by very cold in chilly area, condensed water freezes on the base and may result not to drain. In this case, please remove the bush and the drain pipe at the bottom of unit. (Left and centre near discharge portion of air, each 1 place). It becomes smooth drain. Ensure that the distance from the drain hole to the ground is 250 mm or more.

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INSTALLATION OF REFRIGERATING PIPES AND AIR REMOVAL

1 Preparation of Pipe

- Use a pipe cutter to cut the copper pipe and remove burn.

- Before flaring, please put on the flare nut.

- Please use exclusive tool for refrigerant R410.

⚠ CAUTION

- Remove burr and jagged edge will cause leakage.
- Point the side to be trimmed downwards during trimming to prevent copper chips from entering the pipe.

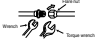
Outer Diameter (in)	A (mm), Rigid Flaring Tool	
	For R410A tool	For R22 tool
6.35 (1/4")	0 – 0.5	1.0
9.35 (3/8")	0 – 0.5	1.0

2 Pipe Connection

⚠ CAUTION

- In case of removing flare nut of a indoor unit, first remove a nut of small diameter side, or a seal cap of big diameter side will fly out.
- Free from water into the piping when working.
- Be sure to tighten the flare nut to the specified torque with a torque wrench. If the flare nut is overtightened, the nut may be split after a long period has passed, and may cause a refrigerant leak.

- Please be careful when bending the copper pipe.
- Screw in manually while adjusting the centre. After that, use a torque wrench to tighten the connection.



	Outer diameter of pipe (in)	Torque N•m (kgf•cm)
Small diameter side	6.35 (1/4")	13.7 – 18.6 (140 – 190)
Large diameter side	9.52 (3/8")	34.3 – 44.1 (350 – 450)
Valve head cap	Small diameter side	6.35 (1/4")	19.6 – 24.5 (200 – 250)
	Large diameter side	9.52 (3/8")	19.6 – 24.5 (200 – 250)
Valve core cap			12.3 – 15.7 (125 – 160)

3 Remove of Air from the Pipe and Gas Leakage Inspection

Procedures of using Vacuum Pump for Air Removal

1. Remove the cap of valve core. Then, connect the charge hose. Remove the cap of valve head. Connect the vacuum pump adapter to the vacuum pump and connect the charge hose to the adapter.

2. Fully tighter the "H" shuttle of the manifold valve and completely unscrew the "Lo" shuttle. Run the vacuum pump for about 10-15 minutes, then completely tighten the "Lo" shuttle and switch off the vacuum pump.
   - Loosen the spindle of the service valve with small diameter by 1/4 turn and tighten the spindle immediately after 5 to 6 seconds.
   - Remove the charging hose from the service valve.

3. Completely unscrew the spindle of the service valve (at 2 places) in anticlockwise direction to allow the flow of refrigerant (using Hexagonal Wrench key).
4. Tighten the cap of valve head. Check the cap's periphery if there is any gas leakage. The task is then completed.

Gas leakage inspection

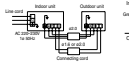
Please use gas leakage detector to check if leakage occurs at connection of Flare nut as shown on the right.

If gas leakage occurs, further tighten the connection to stop leakage.

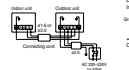
CONNECTION OF POWER CORD

Procedures of Wiring

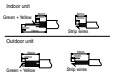
When power supplies to Indoor Unit



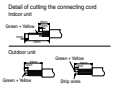
When power supplies to Outdoor Unit



Detail of cutting the connecting cord



Detail of cutting the connecting cord



⚠ WARNING

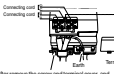
- THIS APPLIANCE MUST BE EARTHED.

⚠ WARNING

- The naked part of the wire core should be 10mm fix it to the terminal tightly. Then try to pull the individual wire to check if the contact is tight. Improper insertion may burn the terminal.
- Be sure to use only wire specified for the use of air-conditioner.
- Please refer to the manual for wire connection and the wiring technique should meet the standard of the electrical installation.
- There is an AC voltage drop between the LN terminal if the power is on. Therefore, be sure to remove the plug from its socket.

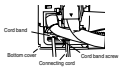
Wiring of the Indoor Unit

- **For wire connection of the indoor unit, you need to remove the front cover (refer to page 14), the low cover from the body of the unit and terminal cover.**
- Remove the cover from the terminal base and screw the cable.



After remove the screw and terminal cover, and put the connecting cords and fix the terminal cover with screw.

- Wiring for the horizontal piping from the right side.



Securely screw in the power cord and connecting cord so that it will not get loose or disconnect.

Tightening torque reference value: 1.2 to 1.6 N-m (12 to 16 kgf-cm).

Excessive tightening may damage the interior of the cord requiring replacement.

When putting two connecting cords through the band.

⚠ WARNING

After wiring the indoor unit, make sure to reattach the terminal cover.

Not to interfere with the drain hose, fold the cable at the drain hose side of the electric box as shown in the left figure, and pass the cable just over the electric box and pull it out.

⚠ WARNING

- Leave some space in the connecting cord for maintenance purpose and be sure to secure it with the cord band.
- Secure the connecting cord along the coated part of the wire using the cord band. Do not exert pressure on the wire as this may cause overheating or fire.

When power supplied to outdoor unit

- Power cannot be turned off using power switch. Always keep the switch "off" and put the switch cover on the power switch so that the power cannot be turned on.
- Power cord is not necessary, store it in the lower-side space on the rear side of indoor unit. When storing, warp the power plug with the vinyl bag for the parts, etc. and seal with adhesive tape to prevent dust or dirt.
- Terminal A and B of indoor unit do not need to be connected.
- When power supplied to outdoor unit, power is not turned off by switching the power of indoor unit to "OFF". In that case, please cut the breaker.

4 Power Source and Operation Test

Power source

⚠ WARNING

- Never extend the power cord.
- Keep additional length for the power cord and do not render the plug under external force as this may cause poor contact.
- Do not fix the power cord with U-shape nail.
- The power cable easily generates heat. Do not bring the cable together with a wire or vinyl tie.

Operation test

- Please ensure that the air conditioner is in normal operating condition during the operation test.
- Explain to your customer the proper operation procedures as described in the user's manual.
- If the indoor unit won't operate, check the cable for correct connection.
- Turn on the lamp in the room where the indoor unit is installed and check the remote controller for normal operation.

After completing the service switch operation, be sure to keep the switch pressed for one or more seconds and stop the force-cooling operation.

Force-cooling Operation

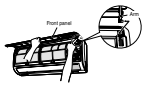
- When the service switch of the outdoor unit is pressed for 1 or more seconds, the force-cooling operation starts. Use this mode when performing the failure diagnosis or collecting refrigerant into the outdoor unit.

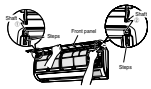
⚠ CAUTION

- Do not operate the unit for more than 5 minutes while the spindle of the service valve is closed.

Removing and Attaching the Front Panel

- Please ensure that the front panel is attached with both hands.





Removing

- After opening the front panel all the way with both hands, press the arm on the far right side to the inside and remove it from the hole, then pull forward.

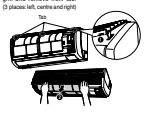
Attaching

- Take both arms of the front panel and press into the holes along the steps of the device. Close the front panel after confirming that the arms are attached.

Attaching the Top Grill

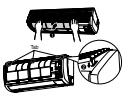
Removing

① Lift up while pressing the top grill and remove from tab. (3 places: left, centre and right)
② Pull forward.



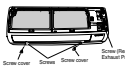
Attaching

① Position the "♡" sign away. Slide along the top of the indoor device and press.
② Press the top grill down and affix with the hold-down tab.

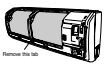


Removing the front cover

① Remove the front panel, pre-filter, air purifying unit, top grill and exhaust pre-filter. (Refer to page 10 of Instruction Manual)
② Remove the screw cover at the bottom of the front cover and remove the screw.
③ Remove the screw behind the exhaust pre-filter.



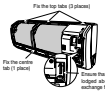
④ Remove the centre tab of the front cover (1 place).



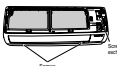
⑤ Place your hands on both sides of the front cover and remove while pulling in the direction of the arrow.

Attaching the front cover

① Confirm that the w-plate has been installed.
② After inserting the front cover into the device, ensure that the top tabs (3 places) have been inserted. In addition, secure the centre tab (1 place). At this time, ensure that the front cover is not lodged above the two way air exchange fan unit.



③ Tighten the screw to the two way air exchange fan unit.
④ Tighten the two screws at the bottom of the front cover and replace the screw cover.



⑤ Replace the exhaust pre-filter, top grill, air purifying unit, pre-filter and front panel.

Wiring of the Outdoor Unit

- Please remove the side cover for wire connection.

⚠ WARNING

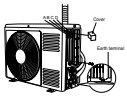
- If you cannot attach the side plate due to the connection cord, please press the connecting cord in the direction to the front panel to fix it.
- Be sure that the hooks of the side cover fixed in certainly. Otherwise water leakage may occur and this causes short circuit or faults.
- The connecting cord should not touch to service valve and pipes. (It becomes high temperature in heating operation.)

Checking for the Electric Source and the Voltage Range

- Before installation, the power source must be checked and necessary wiring work must be completed. To make the wiring capacity proper, use the wire gauge list below for the wiring from house distribution fuse box to the outdoor unit in consideration of the blocked rotor current.

IMPORTANT

Wire length	Wire gauge
up to 6m	1.6mm²
up to 15m	2.5mm²
up to 20m	4.0mm²



- Investigate the power supply capacity and other electrical conditions at the installing location.
- Depending on the model of room air conditioner to be installed, request the customer to make arrangements for the necessary electrical work etc.
- The electrical work includes the wiring work up the outdoor unit. In localities where electrical conditions are poor, use of a voltage regulation is recommended.
- Install outdoor for the room air conditioner within the reaching range of the line cord.

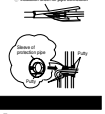
IMPORTANT

Fuse Capacity
16A time delay fuse

FINAL STAGE

1 Heat Insulation and Finish of the Piping

- The connected terminals should be completed sealed with heat insulator and then tied up with rubber strap.
- Do not tie the terminals with the tape too tight.
- If any clearance or over-tightening may cause condensation.
- Please tie the pipe and power cord together with vinyl tape as shown in the figure showing the installation of Indoor and Outdoor units.
- To enhance the heat insulation and to prevent water condensation, please cover the outdoor part of the drain hose and pipe with insulation sheet.
- Completely seal any gap with putty.

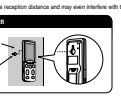


2 Attixing the Remote Controller and Installing the Air Purifying Filter

- The remote controller may be affixed to a wall or pillar with the remote controller mounting screw ④.
- When operating the air conditioning with the remote controller attached, please confirm that the signal is received by the air conditioning. Furthermore, fluorescent lighting may affect the reception of the signal, therefore, please confirm by turning on the fluorescent lighting even during the day.
- Lighting fixtures with electronic starters may shorten the reception distance and may even interfere with the signal being received.

Installation

- Insert the remote controller on the mounting bracket from above.



Address Changing Switch

This is utilized to avoid the interference of remote controller signals when there are two indoor devices installed in the same room. The Address Changing Switch is on the outside of the remote controller battery cover (set to "A" during shipment).

- Setting the address (to avoid interference)

Only one of the two indoor devices has to be set (turn off the power to the other indoor device).

① Place a battery in the remote controller and press the reset switch (please refer to page 29 in the Instruction Manual).
② While pointing the remote controller transmitter/receiver at the indoor device, move the switch of the Address Changing Switch to "B".
③ The channel is set when the device beeps, confirming it has received the signal.

- After changing the address, please confirm the device's operation with the remote controller. If the device does not operate, return the switch to "A" and please set the switch again.

- Installing and verifying the air purifying filter/duct filter (fresh air in pre-filter, exhaust pre-filter, nano-titanium disinfecting deodorizing/dust collecting fresh air in filter). (Refer to Instruction Manual pages 10, 11.)

- Remove the air purifying unit (2 places) by pulling out while pushing up with both hands.

- Open the air purifying unit and attach the air purifying filter with the jagged side up.

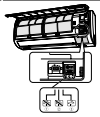
- Remove the fresh air in pre-filter and the filter frame.

- Attach the ⑨ Nano-Titanium disinfecting deodorizing/dust collecting fresh air in filter to the filter frame.

- Attach the filter frame and the fresh air in pre-filter.

- Improper filter installation may result in abnormal noises.

3 Setting the Two Way Air Exchange Selection Switch



Operate Two Way Air Exchange	When operating the Two way air exchange. Please utilize this mode for normal operation.
Only Exhaust	When stopping the Fresh air in operation. Please select this mode when there is space to vent the Two way air exchange hose between the internal and external walls, which leads outside the room, even if the mounting work cannot directly vent the Two way air exchange hose outside the room.
Prohibit Two Way Air Exchange	When stopping the Two way air exchange operation. Please select this mode when the Two way air exchange hose cannot be vented directly outside the room (Installation work specified for the embedded ducting) or when there is a source of a questionable odor, smoke, or oily smoke in the vicinity of the opening of the Two way air exchange hose, even if the mounting work directly vents the Two way air exchange hose outside the room.

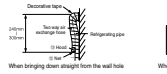
End Treatment of Two Way Air Exchange Hose

- Treat the end of two way air exchange hose placed outdoor as follows.

When not using the outlet cover

① Cut the two way air exchange hose so that the height difference between downward bending point and the tip of the hood becomes 240 – 300mm.

- When the two way air exchange hose can be brought down straight from the wall hole, cut the hose at the point 190 – 250 mm away from the wall hole.
- When arranging the two way air exchange hose in outdoor, cut the hose at the point 190 – 250 mm away from the downward bending point.

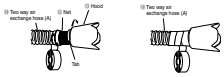


② Connect the two way air exchange hose to the net.

When the two way air exchange hose is ø25

Attach the hood to the net, insert the two way air exchange hose in the net and fix with vinyl tape.

(1) Insert the hood until it touches the end.
(2) Fix the tab of the net to the tab of the hood.
(3) Wrap the tip of the two way air exchange hose with vinyl tape (locally procured) 2 – 3 round.
(4) Insert the two way air exchange hose (A) inside the net until it touches the end and fix it with vinyl tape (locally procured).



③ Fix the two way air exchange hose with the drain hose with decorative tape. (Refer to the above drawing.) In this case, face it downwards to avoid covering the two way air exchange opening.

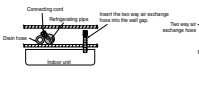
In case a hole cannot be opened in the outer wall

Only the exhaust is possible when the hole in the wall passes through to the outside of the room.

① Install the net to the two way air exchange hose and seal with vinyl tape.
② Place the two way air exchange hose in the space within the wall.
③ Ensure that the two way air exchange Selection Switch is on "Stop Fresh air in" (refer to "Setting the two way air exchange Selection Switch" on page 12.)

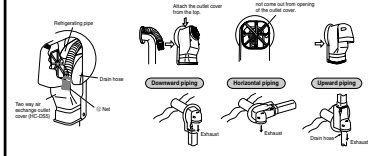
⚠ CAUTION

- Do not supply air from inside a wall. By supplying highly humid air from inside a wall or from underneath the floor, this will cause dew to form. In addition, the odors from inside a wall or from underneath the floor may cause the odors to infiltrate the room.



Using two way air exchange outlet cover (HC-DSS)

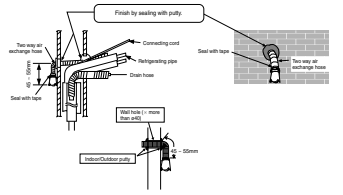
① Cut the two way air exchange hose at 45-55mm from wall.
② Wrap the top of the two way air exchange hose with vinyl tape (procure locally) tightly.
③ Attach the net to the two way air exchange hose, and then seal with vinyl tape (procure locally).
④ Attach the two way air exchange outlet cover.



In Case of Embedded Piping

In case a hole can be opened in the outer wall.

① Open a hole of ø40 and above on the external wall at right bottom side of the air conditioner.
② Pull out the two way air exchange hose, attach the ⑫ net and ⑪ hood, then seal with vinyl tape.



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